Synergism of alcohol, diabetes, and viral hepatitis on the risk of hepatocellular carcinoma in blacks and whites in the U.S.

BACKGROUND: Heavy alcohol consumption, viral hepatitis, and diabetes are risk factors for hepatocellular carcinoma (HCC). However, to the authors' knowledge, the information concerning their interaction effect in patients with risk of HCC is sparse. METHODS: A population-based, case-control study of HCC was conducted during 1984-2002. The study involved 295 HCC cases and 435 age-, gender-, and race-matched control subjects among Hispanic and non-Hispanic whites and blacks in Los Angeles County, California. Lifestyle risk factors were ascertained through in-person interviews. Infections with the hepatitis B and C (HCV) viruses were determined using their serologic markers. RESULTS: Fourteen HCC cases but no control subjects tested positive for the hepatitis B surface antigen. Seropositivity for antibodies to HCV was associated with an odds ratio (OR) of 125 (95% confidence interval [95% CI], 17-909) for HCC, whereas seropositivity for antibodies to the hepatitis B core antigen was related to an OR of 2.9 (95% CI, 1.7-5.0). Heavy alcohol consumption and cigarette smoking were found to be independently associated with a statistically significant two to threefold increase in risk of HCC after adjustment for hepatitis B and C serology. Subjects with a history of diabetes had an OR of 2.7 (95% CI, 1.6-4.3) for HCC compared with nondiabetic subjects. A synergistic interaction on HCC risk was observed between heavy alcohol consumption and diabetes (OR = 4.2; 95% CI, 2.6-5.8), heavy alcohol consumption and viral hepatitis (OR = 5.5; 95% CI, 3.9-7.0), or between diabetes and viral hepatitis (OR = 4.8; 95% CI, 2.7-6.9). CONCLUSIONS: Heavy alcohol consumption, diabetes, and viral hepatitis were found to exert independent and synergistic effects on risk of HCC in U.S. blacks and whites.